Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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EX5

Elgamal CRYPTOGRAPHIC ALGORITHM

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Code: (--CyptoFunctions--)

Cypto_functions.py:

```
import random

def is_prime(n, k=40):
    if n < 2:
        return False
    if n == 2 or n == 3:
        return True
    if n % 2 == 0:
        return False
    r, d = 0, n - 1
    while d % 2 == 0:
        r += 1
        d //= 2
    for _ in range(k):
        a = random.randint(2, n - 2)
        x = pow(a, d, n)
        if x == 1 or x == n - 1:
            continue
        for _ in range(r - 1):
            x = pow(x, 2, n)
            if x == n - 1:
                break
        else:
            return False
    return True
def generate_prime(bits=512):
    while True:
        q = random.getrandbits(bits)
        q |= (1 << bits - 1) | 1
        if is_prime(q):
            return q
def key_generation():
    q = generate_prime(512)
    alpha = 2
    Xa = random.randint(2, q - 2)
    Ya = pow(alpha, Xa, q)
    return q, alpha, Xa, Ya
```

```python
def encrypt(message, q, alpha, Ya):
    M = int.from_bytes(message.encode(), 'big')
    k = random.randint(2, q - 2)
    K = pow(Ya, k, q)
    C1 = pow(alpha, k, q)
    C2 = (K * M) % q
    return k, K, C1, C2

def decrypt(C1, C2, q, Xa):
    K_dec = pow(C1, Xa, q)
    M_dec = (C2 * pow(K_dec, -1, q)) % q
    message_bytes = M_dec.to_bytes((M_dec.bit_length() + 7) // 8, 'big')
    return K_dec, M_dec, message_bytes.decode()
```

Sender:

```
import socket
import crypto_functions as cf

def main():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect(("localhost", 8000))

    data = b''
    while data.count(b'\n') < 3:
        data += client.recv(4096)

    lines = data.decode().strip().split('\n')
    q = int(lines[0])
    alpha = int(lines[1])
    Ya = int(lines[2])

    print("--- ENCRYPTION BY SENDER ---")
    messageStr = input("Enter Plaintext message (M): ")

    k, K, C1, C2 = cf.encrypt(messageStr, q, alpha, Ya)

    print(f"Generated random integer k: {k}")
    print(f"Formula: K = (Ya^k) mod q")
    print(f"Calculated K: {K}")
    print(f"Formula: C1 = (alpha^k) mod q")
    print(f"Calculated C1: {C1}")
    print(f"Formula: C2 = (K * M) mod q")
    print(f"Calculated C2: {C2}")

    client.sendall(str(C1).encode() + b'\n')
    client.sendall(str(C2).encode() + b'\n')

    print("\nCiphertext (C1, C2) sent to Receiver.")

    client.close()

if __name__ == "__main__":
    main()
```

Receiver:

```
import socket
import crypto_functions as cf

def main():
    print("--- KEY GENERATION BY RECEIVER ---")

    q, alpha, Xa, Ya = cf.key_generation()

    print(f"Generated q (prime number): {q}")
    print(f"Generated alpha (primitive root of q): {alpha}")
    print(f"Generated Xa (private key): {Xa}")
    print(f"Formula: Ya = (alpha^Xa) mod q")
    print(f"Calculated Ya (public key): {Ya}")

    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server.bind(("localhost", 8000))
    server.listen(1)

    print("\nWaiting for Sender...")

    conn, addr = server.accept()

    conn.sendall(str(q).encode() + b'\n')
    conn.sendall(str(alpha).encode() + b'\n')
    conn.sendall(str(Ya).encode() + b'\n')

    data = conn.recv(8192).decode().strip().split('\n')
    C1 = int(data[0])
    C2 = int(data[1])
    print("\n--- DECRYPTION BY RECEIVER ---")
    print(f"Received Ciphertext: (C1, C2) = ({C1}, {C2})")
    K_dec, M_dec, message = cf.decrypt(C1, C2, q, Xa)
    print(f"Formula: K = (C1^Xa) mod q")
    print(f"Calculated K: {K_dec}")
    print(f"Formula: M = (C2 * K^-1) mod q")
    print(f"Calculated M (Plaintext): {M_dec}")
    print(f"Final Decrypted Message: {message}")
    conn.close()
    server.close()
if __name__ == "__main__":
    main()
```

Output:

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cry
pto Lab/Ex5$ python receiver.py
--- KEY GENERATION BY RECEIVER ---
Generated q (prime number): 7316474975196443197709541514809316 7
6797865402929799973638869652242655707875604028376567137943302 73
9100329919554320948297798337002836683608189508650176743 7
Generated alpha (primitive root of q): 2
Generated Xa (private key): 7067603342458967705851292470567497 7
4546094264285024718289477735693715784561202064014698340063741 91
0844010439575770524641453400182812620584270715435235895 4
Formula: Ya = (alpha^Xa) mod q
Calculated Ya (public key): 2989904858546191723134350458259589 2
8322700464787233367557276212936996148024485672671698306708904 68
1740145585467216059022122727808253060901302093866474370 5

Waiting for Sender...

--- DECRYPTION BY RECEIVER ---
Received Ciphertext: (C1, C2) = (7780469437792126916113678710 11
0556537437628065590655351484745215443086059233439008017354234 10
6154546147868432859558614125228473656095907673063177846999 83, 3
3080044956846797569050230705258509318281456589149743046422600 14
2988318521542808670067120705190112480181015569925705091529945 99
0311744530124345142983500 22)
Formula: K = (C1^Xa) mod q
Calculated K: 9456819400080069837994663835705767149421351339650
8531231748560130579603326760364998225450925319911585381746825 18
8019528336937914926563398681179313494985 4
Formula: M = (C2 * K^-1) mod q
Calculated M (Plaintext): 310939249775
Final Decrypted Message: Hello
```

```
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cry
pto Lab/Ex5$ python sender.py
--- ENCRYPTION BY SENDER ---
Enter Plaintext message (M): Hello
Generated random integer k: 3310307700209815820499074844255826 9
5177520344809454999546600570904479662499255293373686044363689 95
9457365889260758459543491333212899591566920114343155197 9
Formula: K = (Ya^k) mod q
Calculated K: 9456819400080069837994663835705767149421351339650
8531231748560130579603326760364998225450925319911585381746825 18
8019528336937914926563398681179313494985 4
Formula: C1 = (alpha^k) mod q
Calculated C1: 7780469437792126916113678710110556537437628065 59
0655351484745215443086059233439008017354234106154546147868432 85
9558614125228473656095907673063177846999 83
Formula: C2 = (K * M) mod q
Calculated C2: 3308004495684679756905023070525850931828145658 91
4974304642260014298831852154280867006712070519011248018101556 99
2570509152994599031174453012434514298350022

Ciphertext (C1, C2) sent to Receiver.
(lab) ak@DevLab:~/Desktop/Data/College/Semester6/Assignment/Cry
pto Lab/Ex5$
```